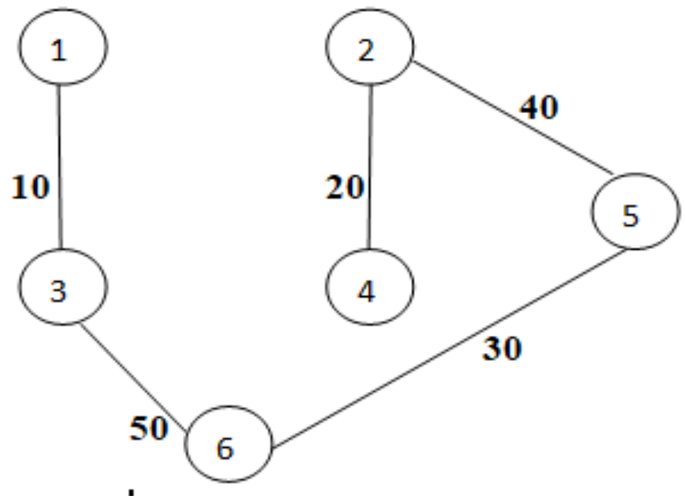
1. Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm.

```
#include<stdio.h>
#include<stdlib.h>
#define MAX 20
int n;                                  // No. of vertices of G
int cost[MAX][MAX];             // Cost matrix
int parent[9];
void ReadMatrix() {
    int i, j;
    printf("Enter the no. of vertices\n");
    scanf("%d",&n);
    printf("Enter the cost adjacency matrix\n");
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n; j++) {
            scanf("%d",&cost[i][j]);
            if (cost[i][j] == 0)
                cost[i][j] = 999;
        }
    }
}
int find(int i) {
    while (parent[i] > 0)
        i = parent[i];
    return i;
}
void uni(int i, int j)
{
    parent[j] = i;
}
```

```
void Kruskals()
 {
    int a = 0, b = 0, u = 0, v = 0, i, j, ne = 1, min, mincost = 0;
    printf("The edges of Minimum Cost Spanning Tree are\n");
    while (ne < n)
{
        for (i = 1, min = 999; i <= n; i++)
{
            for (j = 1; j <= n; j++)
{
                if (cost[i][j] < min)
 {
                    min = cost[i][j];
                    a = u = i;
                    b = v = j;
                }
            }
        }
        u = find(u);
        v = find(v);
        if (u != v)
{
            uni(u, v);
            printf(   " Edge %d:  (%d,%d), Cost = %d\n", ne++, a, b, min);
            mincost += min;
        }
        cost[a][b] = cost[b][a] = 999;
    }
    printf("\nMinimum cost: %d\n", mincost);
}
int main() {
```

```
    printf("*****KRUSKAL'S ALGORITHM*****\n");
    ReadMatrix();
    Kruskals();
    return 0;
}
```



Output:

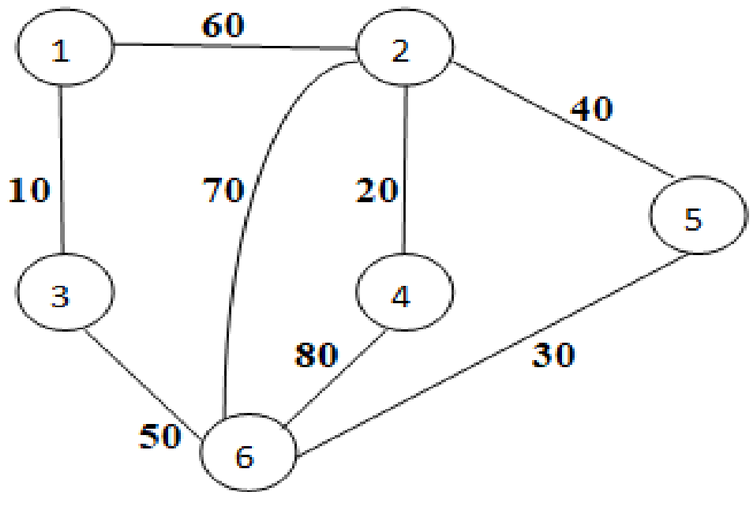
cc 2.c

./a.out

Enter the number of nodes: 6

Enter the adjacency matrix:

0	60	10	999	999	999
60	0	999	20	40	70
10	999	0	999	999	50
999	20	999	0	999	80
999	40	999	999	0	30
999	70	50	80	30	0



The edges of Minimum Cost Spanning Tree are

Spanning Tree:

Edge 1: (1,3) Cost = 10

Edge 2: (2,4) Cost = 20

Edge 3: (5,6) Cost = 30

Edge 4: (2,5) Cost = 40

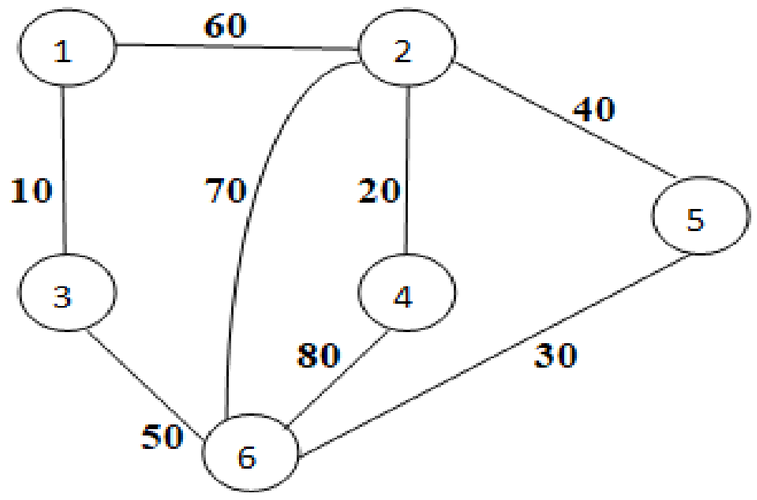
Edge 5: (3,6) Cost = 50

Minimun cost=150

2. Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Prim's algorithm.

```
#include<stdio.h>
int u,v,n,i,j,ne=1;
int visited[10]={0},min,mincost=0,cost[10][10];
void main()
{
        printf("\n Enter the number of nodes:\n");
        scanf("%d",&n);
        printf("\n Enter the adjacency matrix:\n");
        for(i=1;i<=n;i++)
        for(j=1;j<=n;j++)
        {
                scanf("%d",&cost[i][j]);
                if(cost[i][j]==0)
                cost[i][j]=999;
        }
        visited[1]=1;
        printf("\n");
```

```
printf("\nThe edges of Minimum Cost Spanning Tree are\n\n");
while(ne<n)
{
        for(i=1,min=999;i<=n;i++)
        for(j=1;j<=n;j++)
        if(cost[i][j]<min)
        if(visited[i]!=0)
        {
                min=cost[i][j];
                u=i;
                v=j;
        }
    if(visited[u]==0 || visited[v]==0)
    {
    printf("\n Edge %d:(%d %d) cost:%d",ne++,u,v,min);
    mincost+=min;
    visited[v]=1;
    }
    cost[u][v]=cost[v][u]=999;
    }
    printf("\n Minimun cost=%d",mincost);
}
```



Output:

cc 2.c

./a.out

Enter the number of nodes: 6

Enter the adjacency matrix:

0	60	10	999	999	999
60	0	999	20	40	70

10	999	0	999	999	50
999	20	999	0	999	80
999	40	999	999	0	30
999	70	50	80	30	0

The edges of Minimum Cost Spanning Tree are

Edge 1:(1 3) cost:10

Edge 2:(3 6) cost:50

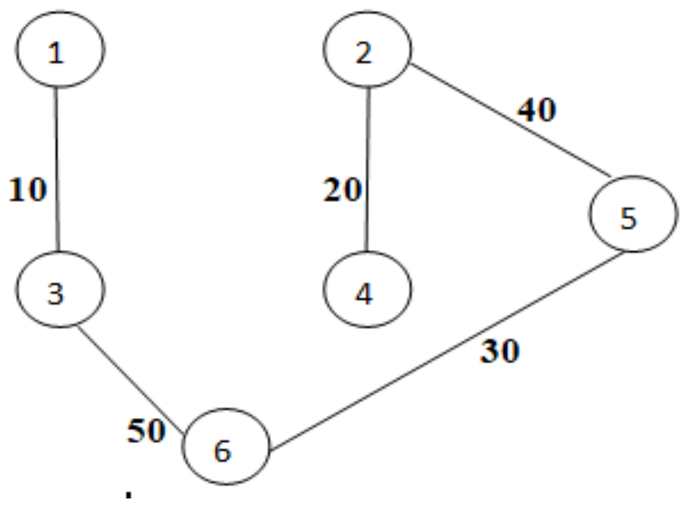
Edge 3:(6 5) cost:30

Edge 4:(5 2) cost:40

Edge 5:(2 4) cost:20

Minimun cost=150

Spanning tree:



3a. Design and implement C/C++ Program to solve All-Pairs Shortest Paths problem using Floyd's algorithm.

```
#include<stdio.h>
#define MAX 20          // max. size of cost matrix
int a[MAX][MAX];        // cost matrix
int n, i, j;            // actual matrix size
void Floyds()
{
```

```
    for (int k = 1; k <= n; k++)
        for (int i = 1; i <= n; i++)
            for (int j = 1; j <= n; j++)
                if ((a[i][k] + a[k][j]) < a[i][j])
                    a[i][j] = a[i][k] + a[k][j];
}
int main()
{
            printf("*****FLOYD'S ALGORITHM*****\n");
            printf("Enter the number of vertices\n");
            scanf("%d",&n);
            printf("Enter the cost matrix of the graph:\n");
            for(i=1;i<=n;i++)
            for(j=1;j<=n;j++)
            scanf("%d",&a[i][j]);
            Floyds();
            printf("All Pair Shortest path matrix\n");
            for(i=1;i<=n;i++)
            {
                        for(j=1;j<=n;j++)
                        printf("%d ",a[i][j]);
                        printf("\n");
                }
}
```

Output:

```
cc 3a.c
./a.out
```

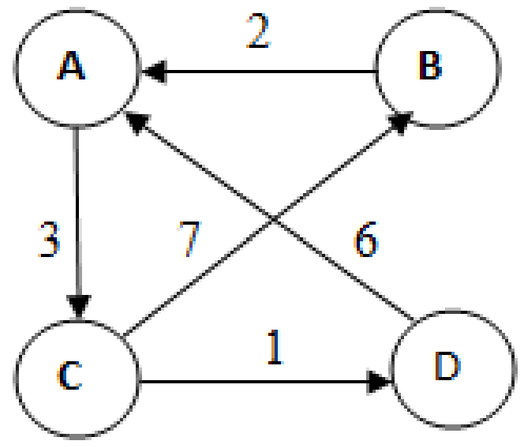
```
*****FLOYD'S ALGORITHM*****
```

Enter the number of vertices

4

Enter the cost matrix of the graph:

0	999	3	999
2	0	999	999
999	7	0	1
6	999	999	0



All Pair Shortest path matrix

0	10	3	4
2	0	5	6
7	7	0	16
6	16	9	0

3b. Design and implement C/C++ Program to find the transitive closure using Warshal's algorithm.

```
#include<stdio.h>
#define MAX 20                          // max. size of matrix
int a[MAX][MAX];                        // adjacency matrix
int n, i, j;                            // actual matrix size
void Warshalls() {
   for (int k = 1; k <= n; k++)
      for (int i = 1; i <= n; i++)
         for (int j = 1; j <= n; j++)
            if (a[i][j] == 1 || (a[i][k] == 1 && a[k][j] == 1))
               a[i][j] = 1;
}
int main() {
   printf("*****WARSHALL'S ALGORITHM*****\n");
   printf("Enter the number of vertices\n");
   scanf("%d",&n);
   printf("Enter the adjacency matrix (1 if there is edge and 0 if there is no direct edge)\n");
   for(i=1;i<=n;i++)
   for(j=1;j<=n;j++)
   scanf("%d",&a[i][j]);
   Warshalls();
   printf("The Transitive Closure Matrix is\n");
   for(i=1;i<=n;i++)
   {
   for(j=1;j<=n;j++)
   printf("%d ",a[i][j]);
   printf("\n");
   }
}
```

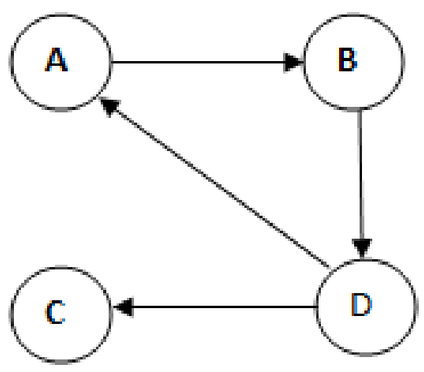
Output:

```
cc 3b.c
./a.out
```

```
*****WARSHALL'S ALGORITHM*****
Enter the number of vertices
4
```

Enter the adjacency matrix (1 if there is edge and 0 if there is no direct edge)

0	1	0	0
0	0	0	1
0	0	0	0
1	0	1	0



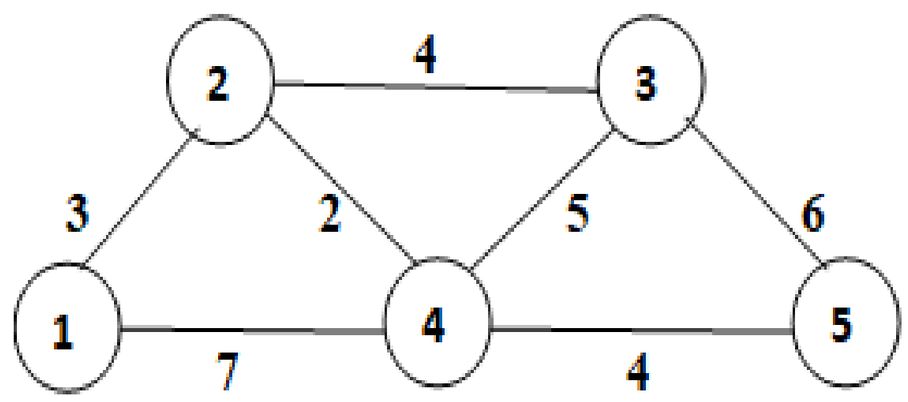
The Transitive Closure Matrix is

1	1	1	1
1	1	1	1
0	0	0	0
1	1	1	1

4. Design and implement C/C++ Program to find shortest paths from a given vertex in a weighted connected graph to other vertices using Dijkstra's algorithm.

```
#include<stdio.h>
#define INF 999
void dijkstra(int c[10][10],int n,int s,int d[10])
{
        int v[10],min,u,i,j;
        for(i=1;i<=n;i++) {
                d[i]=c[s][i];
                v[i]=0;
        }
        v[s]=1;
        for(i=1;i<=n;i++) {
                min=INF;
                for(j=1;j<=n;j++)
        if(v[j]==0&&d[j]<min) {
                min=d[j];
                u=j;
        }
                v[u]=1;
                for(j=1;j<=n;j++)
                if(v[j]==0&&(d[u]+c[u][j])<d[j])
                d[j]=d[u]+c[u][j];
        }
}
int main()
```

```
{
        int c[10][10],d[10],i,j,s,sum,n;
        printf("\nEnter the number of vertices:\n");
        scanf("%d",&n);
        printf("\nEnter the cost adjacency matrix:\n");
        for(i=1;i<=n;i++)
        for(j=1;j<=n;j++)
        scanf("%d",&c[i][j]);
        printf("\nEnter the source vertex:\n");
        scanf("%d",&s);
        dijkstra(c,n,s,d);
        for(i=1;i<=n;i++)
        if(i!=s)
        printf("\nShortest distance from the path %d --> %d is %d",s,i,d[i]);
         return 0;
}
```



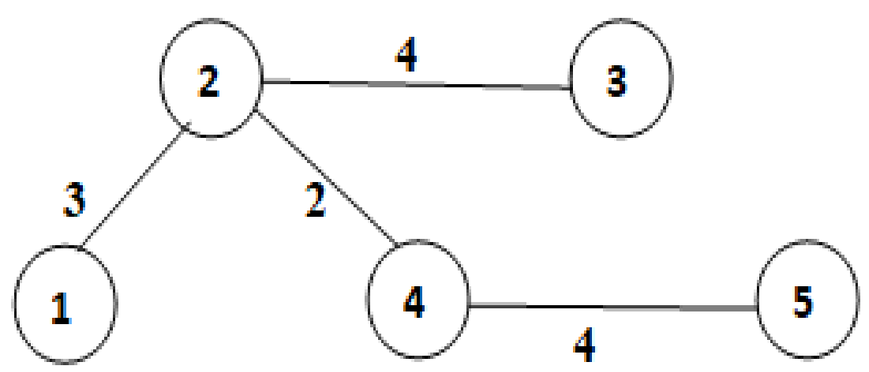
Output:

cc 4.c
./a.out

Enter the number of vertices:

5

Enter the cost adjacency matrix:



0	**3**	**999**	**7**	**999**
3	**0**	**4**	**2**	**999**
999	**4**	**0**	**5**	**6**
7	**2**	**5**	**0**	**4**
999	**999**	**6**	**4**	**0**

Enter the source vertex:

1

Shortest distance from the path 1 -->2 is 3

Shortest distance from the path 1 -->3 is 7

Shortest distance from the path 1 -->4 is 5

Shortest distance from the path 1 -->5 is 9

5. Design and implement C/C++ Program to obtain the Topological ordering of vertices in a given digraph.

```
#include<stdio.h>
int a[10][10],n,indegre[10];
void find_indegre()
{
        int j,i,sum;
       for(j=0;j<n;j++) {
                sum=0;
        for(i=0;i<n;i++)
        sum+=a[i][j];
        indegre[j]=sum;
        }
}
void topology()
{
        int i,u,v,t[10],s[10],top=-1,k=0;
        find_indegre();
```

```
for(i=0;i<n;i++)
{
        if(indegre[i]==0)
        s[++top]=i;
}
while(top!=-1)
{
        u=s[top--];
        t[k++]=u;
for(v=0;v<n;v++)
{
if(a[u][v]==1)
{
        indegre[v]--;
        if(indegre[v]==0)
        s[++top]=v;
}
}
}
printf("The topological Sequence is:\n");
for(i=0;i<n;i++)
printf("%d ",t[i]);
}
void main()
{
        int i,j;
        printf("Enter number of jobs:");
        scanf("%d",&n);
        printf("\nEnter the adjacency matrix:\n");
        for(i=0;i<n;i++)
        {
```

```
        for(j=0;j<n;j++)
        scanf("%d",&a[i][j]);
        }
        topology();
}
```

Output:

cc 5.c

./a.out

Enter number of jobs:6

Enter the adjacency matrix:

0 0 1 1 0 0

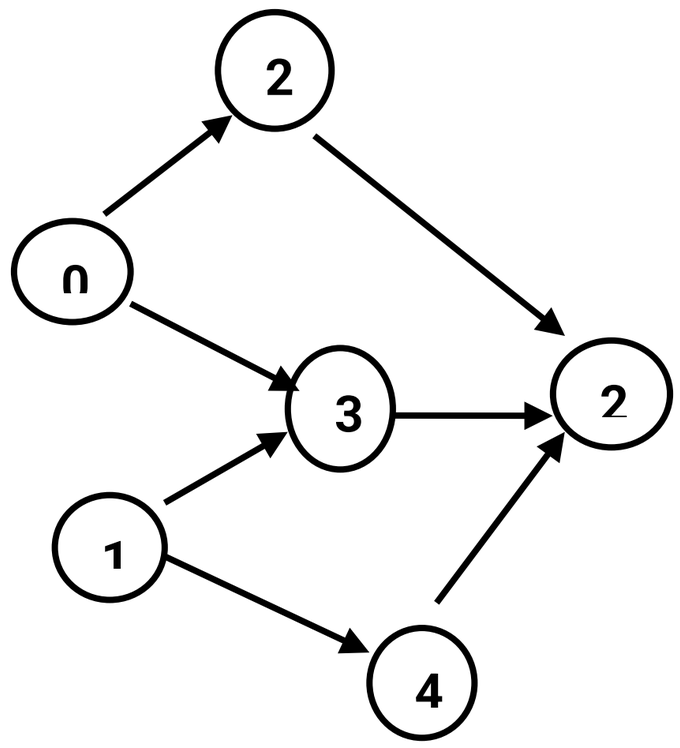
0 0 0 1 1 0

0 0 0 1 0 1

0 0 0 0 0 1

0 0 0 0 0 1

0 0 0 0 0 0



The topological Sequence is:

1 4 0 2 3 5

6. Design and implement C/C++ Program to solve 0/1 Knapsack problem using Dynamic Programming method.

```
#include<stdio.h>
int w[10],p[10],v[10][10],n,i,j,cap,x[10]={0};
int max(int i,int j)
{
        return ((i>j)?i:j);
}
int knap(int i,int j)
```

```
{
        int value;
        if(v[i][j]<0)
        {
                if(j<w[i])
                value=knap(i-1,j);
                else
                value=max(knap(i-1,j),p[i]+knap(i-1,j-w[i]));
                v[i][j]=value;
        }
        return(v[i][j]);
}
void main()
{
        int profit,count=0;
        printf("\nEnter the number of elements\n");
        scanf("%d",&n);
        printf("Enter the profit and weights of the elements\n");
        for(i=1;i<=n;i++)
        {
                printf("For item no %d\n",i);
                scanf("%d\t%d",&p[i],&w[i]);
        }
        printf("\nEnter the capacity \n");
        scanf("%d",&cap);
        for(i=0;i<=n;i++)
        for(j=0;j<=cap;j++)
        if((i==0)||(j==0))
                v[i][j]=0;
                else
                v[i][j]=-1;
```

```
        profit=knap(n,cap);
        i=n;
        j=cap;
while(j!=0&&i!=0)
{
        if(v[i][j]!=v[i-1][j])
        {
                x[i]=1;
                j=j-w[i];
                i--;
        }
        else
        i--;
    }
    printf("\nItems included are\n");
    printf("Sl.no\t weight\t profit\n");
    for(i=1;i<=n;i++)
    if(x[i])
    printf("%d\t\t %d\t\t %d\n",++count,w[i],p[i]);
    printf("Total profit = %d\n",profit);
}
```

Output:

cc 6.c

./a.out

Enter the number of elements

4

Enter the profit and weights of the elements

For item no 1

12 2

For item no 2

10 1

For item no 3

20 3

For item no 4

15 2

Enter the capacity

5

Items included are

Sl.no	weight	profit
1	2	12
2	1	10
3	2	15

Total profit = 37

7. Design and implement C/C++ Program to solve discrete Knapsack and continuous Knapsack problems using greedy approximation method.

```
#include<stdio.h>
#include<stdlib.h>
typedef struct {
    float w;
    float p;
    float r;
} KObject;
int n;                  // no. of objects
float M;        // capacity of Knapsack

void ReadObjects(KObject obj[])
{
    KObject temp;
    printf("Enter the max capacity of knapsack: ");
    scanf("%f",&M);
    printf("Enter Weights: ");
    for (int i = 0; i < n; i++)
        scanf("%f",&obj[i].w);
    printf("Enter Profits: ");
    for (int i = 0; i < n; i++)
        scanf("%f",&obj[i].p);
    for (int i = 0; i < n; i++)
        obj[i].r = obj[i].p / obj[i].w;
    for (int i = 0; i < n - 1; i++)
        for (int j = 0; j < n - 1 - i; j++)
            if (obj[j].r < obj[j + 1].r) {
                temp = obj[j];
```

```
            obj[j] = obj[j + 1];
            obj[j + 1] = temp;
        }
}
void Knapsack(KObject kobj[])
{
    float x[20];
    float totalprofit;
    int i;
    float rc;               // U place holder for M
    rc = M;
    totalprofit = 0;
    for (i = 0; i < n; i++)
        x[i] = 0;
    for (i = 0; i < n; i++) {
        if (kobj[i].w >= rc)
            break;
        else
        {
            x[i] = 1;
            totalprofit = totalprofit + kobj[i].p;
            rc = rc - kobj[i].w;
        }
    }
    if (i < n)
        x[i] = rc / kobj[i].w;
    totalprofit = totalprofit + (x[i] * kobj[i].p);
    printf("The Solution is: \n");
    printf("weight\tprofit\tFraction\t\n");
    for (i = 0; i < n; i++)
        printf("%.2f\t%.2f\t%.2f\n", kobj[i].w, kobj[i].p, x[i]);
```

```
    printf("\nTotal profit is = %.2f\n", totalprofit);
}

int main()
{
    printf("*****KNAPSACK USING GREEDY METHOD*****\n");
    printf("Enter number of objects: ");
    scanf("%d",&n);
    KObject obj[n];
    ReadObjects(obj);
    Knapsack(obj);
    return 0;
}
```

Output:

```
cc 7.c
./a.out
```

```
*****KNAPSACK USING GREEDY METHOD*****
Enter number of objects: 3
Enter the max capacity of knapsack: 40
Enter Weights: 20 25 10
Enter Profits: 30 40 35
The Solution is:
Weight  Profit  Fraction
 10.00   35.00     1.00
 25.00   40.00     1.00
 20.00   30.00     0.25
```

Total profit is = 82.50

8. Design and implement C/C++ Program to find a subset of a given set S = {sl , s2,.....,sn} of n positive integers whose sum is equal to a given positive integer d.

```
#include<stdio.h>
#define MAX 10
int s[MAX],x[MAX],d;
void sumofsub(int p,int k,int r) {
        int i;
        x[k]=1;
        if((p+s[k])==d) {
                for(i=1;i<=k;i++)
                if(x[i]==1)
                printf("%d ",s[i]);
                printf("\n");
}
else
        if(p+s[k]+s[k+1]<=d)
```

```
        sumofsub(p+s[k],k+1,r-s[k]);
        if((p+r-s[k]>=d)&&(p+s[k+1]<=d))
        {
                x[k]=0;
                sumofsub(p,k+1,r-s[k]);
        }
}
int main() {
        int i,n,sum=0;
        printf("Enter the n value:");
        scanf("%d",&n);
        printf("Enter the set in increasing order:");
        for(i=1;i<=n;i++)
        scanf("%d",&s[i]);
        printf("Enter the max subset value:");
        scanf("%d",&d);
        for(i=1;i<=n;i++)
        sum=sum+s[i];
        if(sum<d || s[1]>d)
        printf("No subset possible");
        else
        sumofsub(0,1,sum);
        return 0;
}
```

Output:

cc 8.c

./a.out

1. **Enter the n value: 5**

   Enter the set in increasing order: 1 2 5 6 8

Enter the max subset value: 9

1 2 6

1 8

2. **Enter the n value: 5**

   Enter the set in increasing order: 1 3 4 5 6

   Enter the max subset value: 20

   No subset possible

3. **Enter the n value: 9**

Enter the set in increasing order: 1 2 3 4 5 6 7 8 9

Enter the max subset value: 9

1 2 6

1 3 5

1 8

2 3 4

2 7

3 6

4 5

9

9. Design and implement C/C++ Program to sort a given set of n integer elements using Selection Sort method and compute its time complexity. Run the program for varied values of n>5000 and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.

```
#include<stdio.h>
#include<stdlib.h>
#include<time.h>
void selsort(int a[],int n)
```

```
{
        int i,j,small,pos,temp;
        for(i=0;i<n-1;i++)
        {
                small=a[i];
                pos=i;
        for(j=i+1;j<n;j++)
        {
        if(a[j]<small)
                {
                small=a[j];
                pos=j;
        }
        }
                temp=a[i];
                a[i]=small;
                a[pos]=temp;
        }
}
void main()
{
        int a[10],i,n;
        clock_t start,end;
        printf("\nEnter the n value:");
        scanf("%d",&n);
        for(i=0;i<n;i++)
         a[i] = 5000 + rand() % 999;                    // generate random numbers
         printf("Input Array:\n");
         for (int i = 0; i < n; i++)
         printf("%d\t ", a[i]);
         printf("\n");
```

```
        start=clock();
        selsort(a,n);
        end = clock();
        double time_taken = ((double)(end - start)) / CLOCKS_PER_SEC * 1000; // in
        milliseconds
        printf("\nSorted array is:\n");
        for(i=0;i<n;i++)
        printf("%d\t ",a[i]);
        printf("\nTime taken is:%f",time_taken);
}
```

Output:

```
cc 9.c
./a.out

Enter the n value:5
Input Array:
5478    5664    5153    5268    5500
Sorted array is:
5153    5268    5478    5500    5664
Time taken is:0.002000
```

10. Design and implement C/C++ Program to sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of n>5000 and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.

```
#include<stdio.h>
#include<stdlib.h>
#include<time.h>
#define MAX 10005
int a[MAX];
void QuickSortAlgorithm(int p, int r) {
    int i, j, temp, pivot;
    if (p < r) {
        i = p;
        j = r + 1;
        pivot = a[p];                    // mark first element as pivot
        while (1) {
            i++;
            while (a[i] < pivot && i < r)
                i++;
            j--;
            while (a[j] > pivot)
                j--;
            if (i < j) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            } else
                break;                   // partition is over
        }
```

```
        a[p] = a[j];
        a[j] = pivot;
        QuickSortAlgorithm(p, j - 1);
        QuickSortAlgorithm(j + 1, r);
    }
}
int main() {
    int n;
    printf("***** QUICK SORT***** \n");
    printf("Enter Max array size: ");
    scanf("%d",&n);
    for (int i = 0; i < n; i++)
     a[i] = 5000 + rand() % 999;                    // generate random numbers
    printf("Input Array:\n");
    for (int i = 0; i < n; i++)
     printf("%d\t ", a[i]);
    printf("\n");
    clock_t start = clock();
    QuickSortAlgorithm(0, n - 1);
    clock_t end = clock();
    double time_taken = ((double)(end - start)) / CLOCKS_PER_SEC * 1000; // in milliseconds
    printf("Sorted Array:\n");
    for (int i = 0; i < n; i++)
        printf("%d\t ", a[i]);
    printf("\nTime Complexity in ms for n=%d is: %.6f\n", n, time_taken);
    return 0;
}
```

Output:

```
cc 10.c
./a.out
```

******* QUICK SORT*******

Enter Max array size: 5

Input Array:

5478 5664 5153 5268 5500

Sorted Array:

5153 5268 5478 5500 5664

Time Complexity in ms for n=5 is: 0.001000

11. Design and implement C/C++ Program to sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of n>5000, and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator.

```
#include<stdio.h>
#include<stdlib.h>
#include<time.h>
#define MAX 10005
int a[MAX];
void MergeSort(int low, int high);
void Merge(int low, int mid, int high);
int main() {
 int n;
 printf("***** MERGE SORT***** \n");
 printf("Enter Max array size: ");
 scanf("%d",&n);
 for (int i = 0; i < n; i++)
    a[i] = 5000 + rand() % 999;                    // generate random numbers
    printf("Input Array:\n");
   for (int i = 0; i < n; i++)
```

```
    printf("%d\t ", a[i]);
    printf("\n");
    clock_t start = clock();
    MergeSort(0, n - 1);
    clock_t end = clock();
 double elapsedTime = ((double)(end - start)) / CLOCKS_PER_SEC * 1000;
printf("Sorted Array (Merge Sort):\n");
for (int i = 0; i < n; i++)
 printf("%d\t ", a[i]);
  printf("\n Time Complexity (ms) for n = %d is : %f\n", n, elapsedTime);
  printf("\n");
  return 0;
}

void MergeSort(int low, int high) {
    int mid;
    if (low < high) {
        mid = (low + high) / 2;
        MergeSort(low, mid);
        MergeSort(mid + 1, high);
        Merge(low, mid, high);
    }
}
void Merge(int low, int mid, int high) {
    int b[MAX];
    int i, h, j, k;
    h = i = low;
    j = mid + 1;
    while ((h <= mid) && (j <= high)) {
        if (a[h] < a[j]) {
            b[i++] = a[h++];
```

```
    } else {
        b[i++] = a[j++];
    }
}
if (h > mid) {
    for (k = j; k <= high; k++) {
        b[i++] = a[k];
    }
} else {
    for (k = h; k <= mid; k++) {
        b[i++] = a[k];
    }
}
for (k = low; k <= high; k++) {
    a[k] = b[k];
}
}
```

Output:

cc 11.c

./a.out

******* MERGE SORT*******

Enter Max array size: 5

Input Array:

5478 5664 5153 5268 5500

Sorted Array (Merge Sort):

5153 5268 5478 5500 5664

Time Complexity (ms) for n = 5 is : 0.006000

12. Design and implement C/C++ Program for N Queen's problem using Backtracking.

```
#include<stdio.h>
#include<math.h>
#include<stdlib.h>
int a[30],count=0;
int place(int pos)
{
        int i;
        for(i=1;i<pos;i++)
```

```
        {
                if((a[i]==a[pos])||((abs(a[i]-a[pos])==abs(i-pos))))
                return 0;
        }
        return 1;
}
void print_sol(int n)
{
        int i,j;
count++;
printf("\n\nSolution #%d:\n",count);
for(i=1;i<=n;i++)
{
        for(j=1;j<=n;j++)
        {
                if(a[i]==j)
                printf("Q\t");
                else
                printf("*\t");
        }
        printf("\n");
}
}


void queen(int n)
{
        int k=1;
        a[k]=0;
        while(k!=0)
        {
```

```
a[k]=a[k]+1;
while((a[k]<=n)&&!place(k))
a[k]++;
if(a[k]<=n)
{
        if(k==n)
        print_sol(n);
        else
        {
                k++;
                a[k]=0;
        }
    }
    else
    k--;
  }
}
void main()
{
    int i,n;
    printf("Enter the number of Queens\n");
    scanf("%d",&n);
    queen(n);
    printf("\nTotal solutions=%d",count);
}
```

Output:

cc 12.c

./a.out

Enter the number of Queens

4

Solution #1:

*	Q	*	*
*	*	*	Q
Q	*	*	*
*	*	Q	*

Solution #2:

*	*	Q	*
Q	*	*	*
*	*	*	Q
*	Q	*	*

Total solutions=2